SOFTWARE TEST PLAN (STP)

Collaborative Study Material & Quiz Platform

Team Members

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Revision History

Version	Date	Author(s)	Change Summary
1.0	15-09-2026	Neha Prabhu, Nisarga, Pradhyumn Anand, Neeraj R Rugi	Initial Test Plan drafted for Collaborative Study Material & Quiz Platform

1. Introduction

1.1 Purpose

This document defines the test plan for the Collaborative Study Material & Quiz Platform v1.0. It outlines the testing objectives, scope, strategy, resources, schedule and responsibilities required to verify that the system meets the functional, non-functional and security requirements defined in the SRS.

1.2 Scope

Testing covers user authentication and role-based access, study-material upload/search/access, question-bank management, quiz creation and publication, quiz attempts with automatic evaluation, result history and performance analysis, and administrative user/content management. Online proctoring, live video classes and payment processing are excluded from this test plan's scope, consistent with the SRS.

1.3 References

- Collaborative Study Material & Quiz Platform — Software Requirements Specification (SRS) v1.0
- IEEE 829 Standard for Software Test Documentation
- OWASP Top 10 (for security test design)

1.4 Definitions

- SRS — Software Requirements Specification
- RTM — Requirements Traceability Matrix
- STP — Software Test Plan
- UAT — User Acceptance Testing
- RBAC — Role-Based Access Control
- NFR — Non-Functional Requirement

2. Test Items

- Authentication & User Management module
- Study Material Management module
- Question Bank & Quiz Management module
- Results & Performance Analysis module
- Administration module

3. Features to be Tested

Features mapped to SRS requirement IDs:

Req ID	Feature / Requirement
CSMQ-F-001	Registered user login with valid credentials
CSMQ-F-002	User logout and session termination
CSMQ-F-003	Role assignment and restriction of protected functions
CSMQ-F-004	Study material upload with title, category and description
CSMQ-F-005	Browse and search study materials
CSMQ-F-006	Update or remove owned study materials
CSMQ-F-007	View or download authorized study materials

CSMQ-F-008	Create questions with text, options, answer and topic
CSMQ-F-009	Edit or delete owned questions
CSMQ-F-010	Create a quiz from the question bank
CSMQ-F-011	Publish or unpublish a quiz
CSMQ-F-012	Attempt and submit a quiz
CSMQ-F-013	Automatic evaluation and scoring of quiz attempts
CSMQ-F-014	Storage of completed quiz attempt details
CSMQ-F-015	View quiz score and performance history
CSMQ-F-016	Performance summaries (average, quiz-wise, topic-wise)
CSMQ-F-017	Administrator activation/deactivation of user accounts
CSMQ-F-018	Administrator review/removal of inappropriate content
CSMQ-NF-001	Response time for common operations (login, search, results)
CSMQ-NF-002	Data consistency across questions, quizzes, attempts and results
CSMQ-NF-003	Usability of navigation and first-time-user workflows
CSMQ-NF-004	Modular organization of application components
CSMQ-NF-005	Compatibility across the two most recent major browser versions
CSMQ-NF-006	Availability during normal operating periods
CSMQ-SR-001	Authentication required for protected pages/operations
CSMQ-SR-002	Role-based authorization for Student/Contributor/Administrator
CSMQ-SR-003	Secure one-way password hashing (no plaintext storage)
CSMQ-SR-004	HTTPS enforced for client-server communication in production
CSMQ-SR-005	Validation of uploaded files and user-provided inputs

4. Features Not to be Tested

- Online proctoring during quiz attempts (explicitly excluded from SRS scope)
- Live video classes / conferencing functionality (out of scope per SRS)
- Payment processing (out of scope per SRS)
- Internal behaviour of the underlying database engine or file-storage service (assumed reliable; only the application's use of them is tested)
- Native rendering behaviour of third-party desktop/mobile browsers
- Underlying network, hosting, or server-provisioning infrastructure
- UML modelling/diagramming tools used to produce design artefacts

5. Test Approach / Strategy

Levels

- Unit tests (module level, e.g. Authentication, Material, Question Bank, Quiz, Results services)
- Integration tests (application server ⇔ database, module ⇔ module interactions)
- System tests (end-to-end platform workflows across roles)
- Acceptance tests (UAT with representative Student, Contributor and Administrator scenarios)

Types

- Functional testing (core platform features)
- Regression testing (after each build/change)
- Performance testing (response time under normal load)
- Security testing (authentication, authorization, input validation)
- Usability testing (navigation clarity, first-time-user experience)

- Compatibility testing (supported browser versions)

Entry Criteria

A stable build is delivered; the test environment (application server and database) is provisioned and reachable; test data (seed accounts, sample materials and question banks) is available.

Exit Criteria

100% of planned test cases executed; zero critical or high-severity defects open; all acceptance criteria in the SRS satisfied; NFR and security targets met or formally waived.

5.1 Security Validation

- Verify unauthenticated users are rejected or redirected to login on protected pages/operations
- Verify role-based authorization is enforced for Student, Contributor and Administrator operations
- Verify passwords are stored using a secure one-way hash and never in plaintext or logs
- Verify HTTPS is enforced for client-server communication in production
- Validate uploaded files and user-provided inputs (type, size, format) before storage or processing
- Fuzz-test registration, material-upload and question-creation fields for injection and malformed-input handling
- Verify session termination on logout invalidates further access to protected pages

6. Test Environment

Hardware

Standard developer workstations/laptops for server and client testing; representative desktop and mobile devices for compatibility and usability checks.

Software

Python-based application server (framework-agnostic), a relational database for persistent data, and the two most recent versions of major desktop browsers, consistent with the SRS's operating-environment and compatibility requirements.

Tools (indicative, not prescriptive)

- Automated UI/functional testing framework (e.g., Selenium or Playwright)
- API testing tool (e.g., Postman) for backend endpoint verification
- Load/performance testing tool (e.g., JMeter or Locust) for response-time checks
- Defect-tracking tool (e.g., Jira or an equivalent issue tracker)

The tools above illustrate one possible toolchain; equivalent tools may be substituted without changing the test approach, items, or acceptance criteria in this plan.

Test Data

Dummy user accounts across all three roles, seeded study materials, a seeded question bank, and pre-configured quizzes with known answer keys.

7. Test Schedule

Milestones:

Milestone	Date
Test case design	16-september-2026
Test environment setup	38-september-2026
Test execution start	1-october-2026
Test execution end	1-november-2026
UAT	02-november-2026 to 05-november-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual and/or automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Requirements Traceability Matrix
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Neha	Prepare and maintain the test plan, coordinate execution, own security validation sign-off
Test Engineer	Nisarga	Design and execute functional and integration test cases (Auth, Materials, Question Bank, Quiz), log defects
Test Engineer	Pradhyumn	Design and execute results/performance/NFR and security test cases, maintain the RTM, log defects
Developer	All team Members	Support defect triage, fixes and re-testing
Product Owner	Neeraj	Approve test results and sign off on release readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke-test builds from the development team
Test environment downtime (application server/database)	Maintain a backup environment on a secondary host/VM
Incomplete or inconsistent question/answer configuration affecting scoring accuracy	Validate question-bank data through QB test cases before quiz creation and attempt testing begins
Large or malformed material uploads affecting storage or performance	Enforce and test file type/size validation early in the cycle
Ambiguity in role-based access rules across Student/Contributor/Administrator	Confirm the RBAC matrix with the Product Owner before designing security test cases

11. Assumptions & Dependencies

- The application server and database environment will be stable and available for the scheduled test window
- Test data (seed accounts, sample materials, question banks) will be provided or seeded before execution begins
- SRS acceptance criteria remain unchanged during the test cycle
- Test accounts covering Student, Contributor and Administrator roles will be made available

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than 4 hours
- The build is unstable enough to block more than 30% of planned test cases

Resume testing if:

- Blocking defects are resolved
- The environment is confirmed stable

13. Test Case Management & Traceability

13.1 Test Cases

A minimum set of 27 test cases is defined below, covering all functional requirements (authentication, materials, question bank, quiz, results, administration), non-functional requirements (performance, data integrity, usability, compatibility) and security requirements (authentication, RBAC, password hashing, HTTPS, input validation).

TC ID	Req ID	Description	Preconditions	Steps	Expected Result	Priority
TC-Auth-01	CSMQ-F-001	Login with valid credentials	Registered account exists	Submit login form with correct email/username and password	An authenticated session is created and the user reaches their role dashboard	High
TC-Auth-02	CSMQ-F-002	Logout terminates session	User is logged in	Trigger logout, then attempt to reload a protected page	Session is terminated; protected pages redirect to login	Medium
TC-Auth-03	CSMQ-F-003	Role restricts protected functions	Accounts exist for Student, Contributor and Administrator	Log in as a Student and attempt to access a Contributor/Admin-only function	Access is denied for the Student; authorized roles retain access to their permitted functions	High
TC-Mat-01	CSMQ-F-004	Upload study material	Logged in as an authorized Contributor	Upload a material with title, subject/category and description	Material is stored and appears in the contributor's material list	High
TC-Mat-02	CSMQ-F-005	Search study materials	At least one accessible material exists	Search using title, subject or category filters	Only matching, accessible materials are returned	High
TC-Mat-03	CSMQ-F-006	Update/remove owned material	A material owned by the logged-in	Edit the material's	Changes are reflected in the	Medium

			contributor exists	fields, save, then remove the material	library; removed material is no longer available to users	
TC-Mat-04	CSMQ-F-007	View/download authorized material	A permitted and a restricted material both exist	Attempt to open/download each material as the current user	Permitted material opens/downloads successfully; unauthorized access is denied	High
TC-QB-01	CSMQ-F-008	Create a question	Logged in as an authorized contributor	Submit a question with text, options, correct answer and topic	The question is stored and retrievable from the question bank	High
TC-QB-02	CSMQ-F-009	Edit/delete owned question	A question owned by the contributor exists	Edit the question, save, then delete a separate owned question	Updated data is saved; deleted question is unavailable for new quizzes	Medium
TC-Quiz-01	CSMQ-F-010	Create a quiz	Question bank has enough questions	Select questions from the bank and define a title and basic instructions	A quiz is created containing the selected questions and metadata	High
TC-Quiz-02	CSMQ-F-011	Publish/unpublish a quiz	A draft quiz exists (TC-Quiz-01 passed)	Publish the quiz, verify student visibility, then unpublish it	Published quizzes are available to permitted students; unpublished quizzes are not available for new attempts	Medium
TC-Quiz-03	CSMQ-F-012	Attempt and submit a quiz	A published quiz is available to the student	View the quiz questions and submit selected answers	The attempt records the selected answers	High

					and submission time	
TC-Result-01	CSMQ-F-013	Automatic scoring of an attempt	A submitted quiz attempt exists (TC-Quiz-03 passed)	Trigger evaluation of the submitted attempt	Score equals the number of correct answers per the configured answer key	High
TC-Result-02	CSMQ-F-014	Store completed attempt details	An attempt has been scored	Query the student's result history for the attempt	A completed attempt record (student, quiz, score, submission details) is present	High
TC-Result-03	CSMQ-F-015	View score and performance history	Student has one or more completed attempts	Open the results/performance dashboard	Dashboard shows previous attempts and corresponding scores	Medium
TC-Perf-01	CSMQ-F-016	Performance summaries by quiz and topic	Result data exists across multiple quizzes/topics	Open the performance summary view	Average, quiz-wise and topic-wise values are calculated correctly from stored attempts	Medium
TC-Admin-01	CSMQ-F-017	Activate/deactivate a user account	Administrator is logged in; a target account exists	Deactivate the account, attempt login as that user, then reactivate it	Account status changes are enforced on subsequent login/access attempts	Medium
TC-Admin-02	CSMQ-F-018	Review/remove inappropriate content	Administrator is logged in; flagged content exists	Review the content and remove it	Removed content is no longer accessible through normal library/search functions	Medium

TC-Perf-Resp-01	CSMQ-NF-001	Response time under normal load	Test load-generation tool configured	Execute login, material search and result-viewing requests under normal simulated load	90th percentile response time is ≤ 3 seconds	High
TC-INT-01	CSMQ-NF-002	Data consistency across the workflow	A quiz has been attempted and scored	Compare stored questions, quiz configuration, attempt answers and calculated result	Stored result matches submitted answers and configured answer keys in all cases	High
TC-UX-01	CSMQ-NF-003	Usability walkthrough	A first-time test user account is available	Ask a first-time user to locate a material and attempt an available quiz without guidance	User completes both tasks unaided, confirming clear navigation and layout	Medium
TC-Compat-01	CSMQ-NF-005	Cross-browser compatibility	Two most recent versions of major desktop browsers available	Execute core workflows (login, material access, quiz attempt) on each browser	Core workflows function correctly on all supported browsers	Medium
TC-Sec-01	CSMQ-SR-001	Unauthenticated access is rejected	No active session	Attempt to open a protected page/operation without logging in	Access is rejected or the user is redirected to login	High
TC-Sec-02	CSMQ-SR-002	Role-based authorization is enforced	Accounts exist for each role	Attempt operations outside each role's permitted scope	Unauthorized role attempts are denied for every role tested	High
TC-Sec-03	CSMQ-SR-003	Passwords are stored hashed	A user account has been registered	Inspect the stored credential	Password is stored only as a salted hash;	High

				record for that account	plaintext is never present	
TC-Sec-04	CSMQ-SR-004	HTTPS is enforced in production	Production-equivalent environment configured	Attempt a plaintext (HTTP) connection to the application	Application traffic is served over HTTPS; plaintext requests are refused or redirected	High
TC-Sec-05	CSMQ-SR-005	Upload and input validation	Logged in as an authorized user	Submit invalid file types, oversized files and malformed input fields	Invalid types, oversized files and invalid fields are rejected safely with a clear error	High

13.2 Requirements Traceability Matrix (RTM)

The RTM ensures every SRS requirement maps to at least one executable test case or review activity.

Req ID	Requirement	Test Case(s)
CSMQ-F-001	Registered user login	TC-Auth-01
CSMQ-F-002	User logout / session termination	TC-Auth-02
CSMQ-F-003	Role-based restriction of functions	TC-Auth-03
CSMQ-F-004	Study material upload	TC-Mat-01
CSMQ-F-005	Study material search	TC-Mat-02
CSMQ-F-006	Update/remove owned material	TC-Mat-03
CSMQ-F-007	View/download authorized material	TC-Mat-04
CSMQ-F-008	Create question	TC-QB-01
CSMQ-F-009	Edit/delete owned question	TC-QB-02
CSMQ-F-010	Create quiz from question bank	TC-Quiz-01
CSMQ-F-011	Publish/unpublish quiz	TC-Quiz-02
CSMQ-F-012	Attempt and submit quiz	TC-Quiz-03
CSMQ-F-013	Automatic scoring	TC-Result-01
CSMQ-F-014	Store completed attempt	TC-Result-02

Req ID	Requirement	Test Case(s)
CSMQ-F-015	View score/performance history	TC-Result-03
CSMQ-F-016	Performance summaries	TC-Perf-01
CSMQ-F-017	Activate/deactivate user account	TC-Admin-01
CSMQ-F-018	Review/remove inappropriate content	TC-Admin-02
CSMQ-NF-001	Response time ≤ 3s (90th pct)	TC-Perf-Resp-01
CSMQ-NF-002	Data consistency	TC-INT-01
CSMQ-NF-003	Usability / first-time navigation	TC-UX-01
CSMQ-NF-004	Modular component organization	Code Review
CSMQ-NF-005	Browser compatibility	TC-Compat-01
CSMQ-NF-006	Availability during normal operation	Operations Review
CSMQ-SR-001	Authentication required	TC-Sec-01
CSMQ-SR-002	Role-based authorization enforced	TC-Sec-02
CSMQ-SR-003	Passwords hashed, not plaintext	TC-Sec-03
CSMQ-SR-004	HTTPS enforced in production	TC-Sec-04
CSMQ-SR-005	Upload/input validation	TC-Sec-05

14. Test Metrics & Reporting

Metrics collected: % test cases executed, % passed/failed, defect density, defect aging, requirement coverage.

Reports: daily execution status report; final Test Summary Report at the end of the execution cycle.

15. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Product Owner		